Non-vanishment of Trigonometric Polynomials

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Any nonzero trigonometric polynomial P cannot vanish in an entire interval

Let's do induction on N, where N is the degree of the trigonometric polynomial P. For $N = 0$, we have $P(x) = a_0$. Since P is nonzero, $a_0 \neq 0$ and therefore P do not vanish anywhere.

Suppose this is true for every value of N less than k. Let $P(x) = \sum_{|n|\leq k} a_n e^{inx}$. Then suppose there is $a < b$ such that $P|_{(a,b)} \equiv 0$. Let $P|_{(a,b)} = F$. We must have

$$F''(x) = \sum_{|n|\leq k} a_n (in)^2 e^{inx} = -\sum_{|n|\leq k} a_n n^2 e^{inx} = 0 \quad \forall x \in (a,b)$$

Thus

$$k^2 F(x) + F''(x) = \sum_{|n|<k} a_n (k^2 - n^2) e^{inx} = 0 \quad \forall x \in (a,b)$$

Notice that $k^2F + F''$ is a trigonometric polynomial of degree at most $k-1$. By our induction hypothesis, we must have that $k^2F + F''$ is trivial. That is, $a_n(k^2 - n^2) = 0 \quad \forall |n| < k$. Therefore $a_n = 0 \quad \forall |n| < k$. So we remain with $F(x) = a_{-k}e^{-ikx} + a_k e^{ikx} = 0 \quad \forall x \in (a,b)$. By now, it is clear that we must have $a_{-k} = a_k = 0$. HEnce P is trivial. Therefore, our induction proof is complete.